

The modulus function

Lessons 15–16 · Algebra 11, §1.5 (extension) · P1 · 1.6 · P2 · 1.1–1.3 · 28 problems

A modulus equation splits into two cases; a modulus inequality is a region. Check every root against the original equation — squaring and case-splitting both invent answers.

NAME _____

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Easy 7

1 $|7 - 12|$

2 $|-3| + |-4|$

3 Solve $|x| = 11$

4 Solve $|x + 5| = 2$

5 Solve $|x| \leq 4$

6 Solve $|x| > 7$

7 Give the vertex of $y = |x + 3| - 2$

Medium 7

8 Solve $|3x - 2| = 10$

9 Solve $|5 - 2x| = 1$

10 Solve $|x - 4| \leq 6$

11 Solve $|4x + 1| \geq 9$

12 Solve $|x + 2| = |3x - 4|$

13 Give the vertex and the y-intercept of $y = |2x - 6|$

14 Give the range of $y = 5 - |x - 1|$

Hard 7

15 Solve $|x + 1| = 3x - 5$

16 Solve $|x^2 - 9| = 7$

17 Solve $|x - 5| + |x + 2| = 9$

18 Solve $|2x - 1| < |x + 3|$

19 Solve $|x| - x = 8$

20 The graph of $y = |x - a|$ passes through $(1, 4)$ and $(7, 2)$. Find a .

21 Solve $|x^2 - 2x| = x$

Very hard 7

22 Show that $|a| + |b| \geq |a + b|$, and say when the two sides are equal.

23 Solve $|x - 1| + |x - 3| < 4$.

24 Find the greatest and least values of $y = |x - 2| + |x + 4|$.

25 Solve $|x + 3| \geq |2x|$.

26 Explain why $y = |x^2 - 1|$ has no derivative at $x = 1$ but does have one at $x = 0$.

27 $|x - 2| = k$ has exactly two solutions. Find the values of k .

28 Sketch $y = |x| - |x - 4|$ and give its range.
